Mesh Radio v1 (working name) — OffGrid Devices
LoRa 902–928 MHz mesh radio, up to ~3 W; Bluetooth LE; one LiPo cell; USB-C, solar and drone power.
NDAA-ready parts list: every part from a US / EU / Japan / Taiwan / Korea / Norway maker.
Each box below is one page. Every connection is a named label; the same name is the same wire.
This drawing is generated from src/circuit.py by src/sch.py: change the code, not the drawing.

Power in: USB-C, solar, drone 5 V (each surge-clamped and fused)

Every input: 22 V flat-clamp TVS (solar and drone behind a reverse-blocking Schottky), then a TPS259470A eFuse. Battery fully isolated while the eFuse is off.

File: inputs.kicad_sch

Charger and battery: protection, reverse polarity, battery sense

Battery fully isolated while the eFuse is off. Charge 1.5 A, solar 1.0 A, 10 V/10 s timer. All current from the USB side (OCD ~4.1 A, SCD ~21 A).

File: charger.kicad_sch

System power: on/off switch, 3.3 V rail

The 3.3 V rail is reverse-battery protected. Buck 2.5 A, 1 MHz, with PFM for light loads.

File: system.kicad_sch

Processor and Bluetooth: nRF52840

nRF52840 in normal-voltage mode: VDD = VDDH = the 3.3 V rail, so its pins and the SX1262 share one logic level.

File: mcu.kicad_sch

LoRa radio: SX1262, its clock, transmit match and receive balun

Semtech SX1262MB2CAS (915 MHz) reference values. TCXO on DIO3 (set 2.2 V in firmware), DIO2 drives the antenna switch.

File: radio.kicad_sch

RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA

Drive SX1262 +11 dBm -> 30 dBm (1 W), +17 dBm -> ~35 dBm (3 W). The 15.9 dB pad keeps the amplifier input at +11 dBm when the battery is at its highest (charge 4.2 V, only ~6 dB to lose).

File: frontend.kicad_sch

Amplifier power: battery-first supply, buck-boost, control voltage, temperature

Amplifier runs from the battery through a 1 Ω sense resistor (charge 2.5 A only when the battery), then a TPS63810 buck-boost set over the SX1262 maximum (range 2.7 V to 5.5 V, 3.6 V typ).

File: pa_power.kicad_sch

Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, buttons, LEDs

Drone connector, 6 pin, 1.25 mm pitch, for the Holybro Micro M10 GPS (pins numbered 1 to 6 as printed on the connector): 1 5V, 2 TX, 3 RX, 4 CTS, 5 RTS, 6 GND. Series resistors on the UART lines.

File: ports.kicad_sch

Generated from src/circuit.py by src/sch.py. Edit the circuit, not this file.

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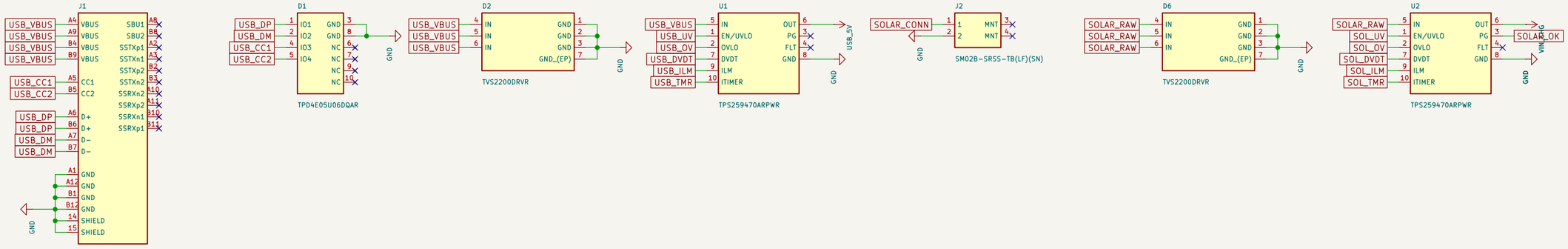
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File: mesh-radio.kicad_sch

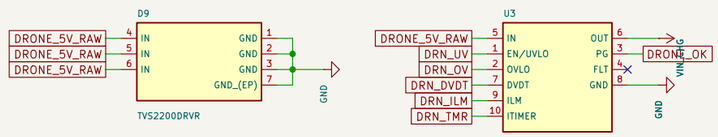
Title: Mesh Radio v1 (working name)

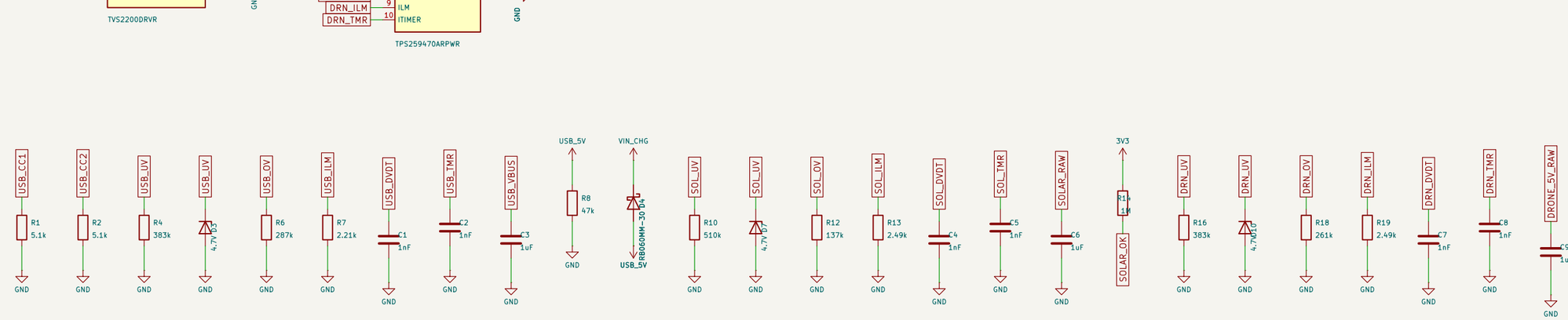
Size: A3	Date:	**Rev: v1**
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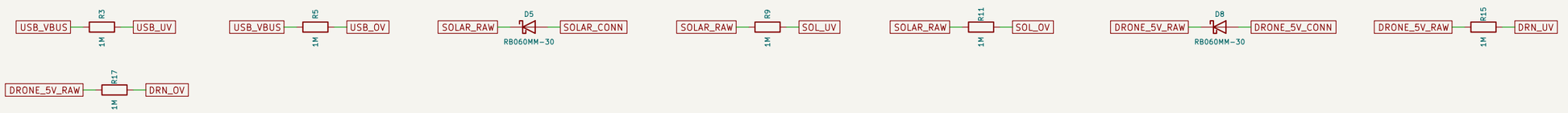
Power in: USB-C, solar, drone 5 V (each surge-clamped and fused)

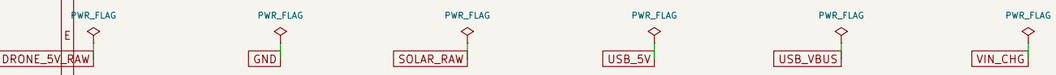
Every input: 22 V flat-clamp TVS (solar and drone behind a reverse-blocking Schottky), then a TPS259470A eFuse. Cut-offs: USB 5.4 V / 1.5 A, drone 5.8 V / 1.3 A, solar 10 V / 1.3 A. All survive 20 V.











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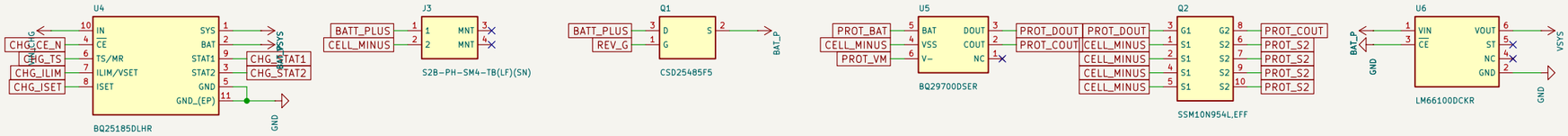
Sheet: /Power in: USB-C, solar, drone 5 V (each surge-clamped and fused)/
File: inputs.kicad_sch

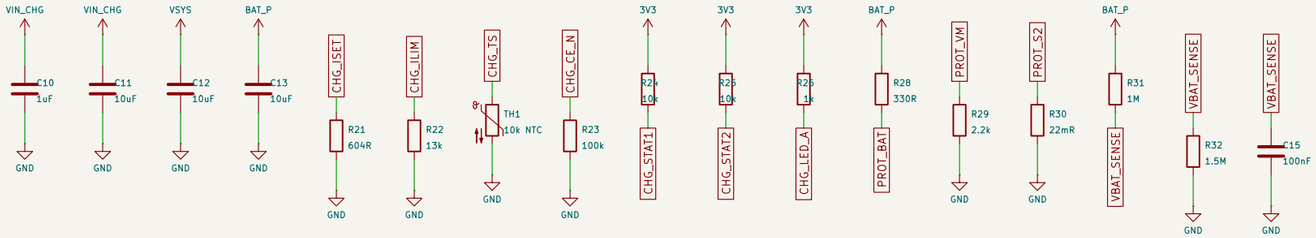
Title: Power in: USB-C, solar, drone 5 V (each surge-clamped and fused)

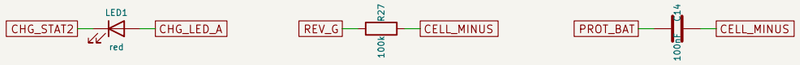
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Charger and battery: protection, reverse polarity, battery sense

Battery plug wired the Adafruit/SparkFun way (pad 1 = +). Cell protection on the minus side
(OCD ~4.1 A, SCD ~21 A, OV 4.275 V, UV 2.8 V); reverse-battery P-FET on the plus side.









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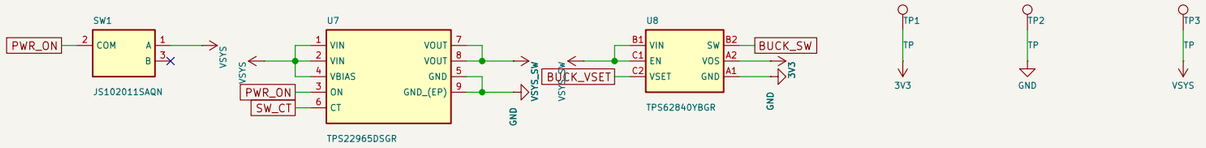
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File: charger.kicad_sch

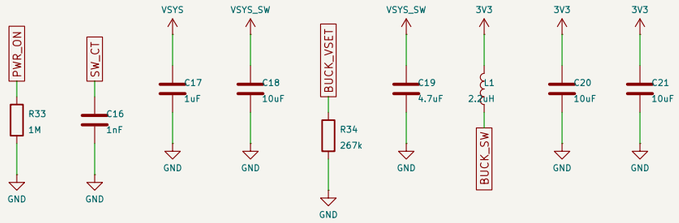
Title: Charger and battery: protection, reverse polarity, battery sense

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System power: on/off switch, 3.3 V rail

The slide switch only drives the load switch enable; charging works with the board switched off.





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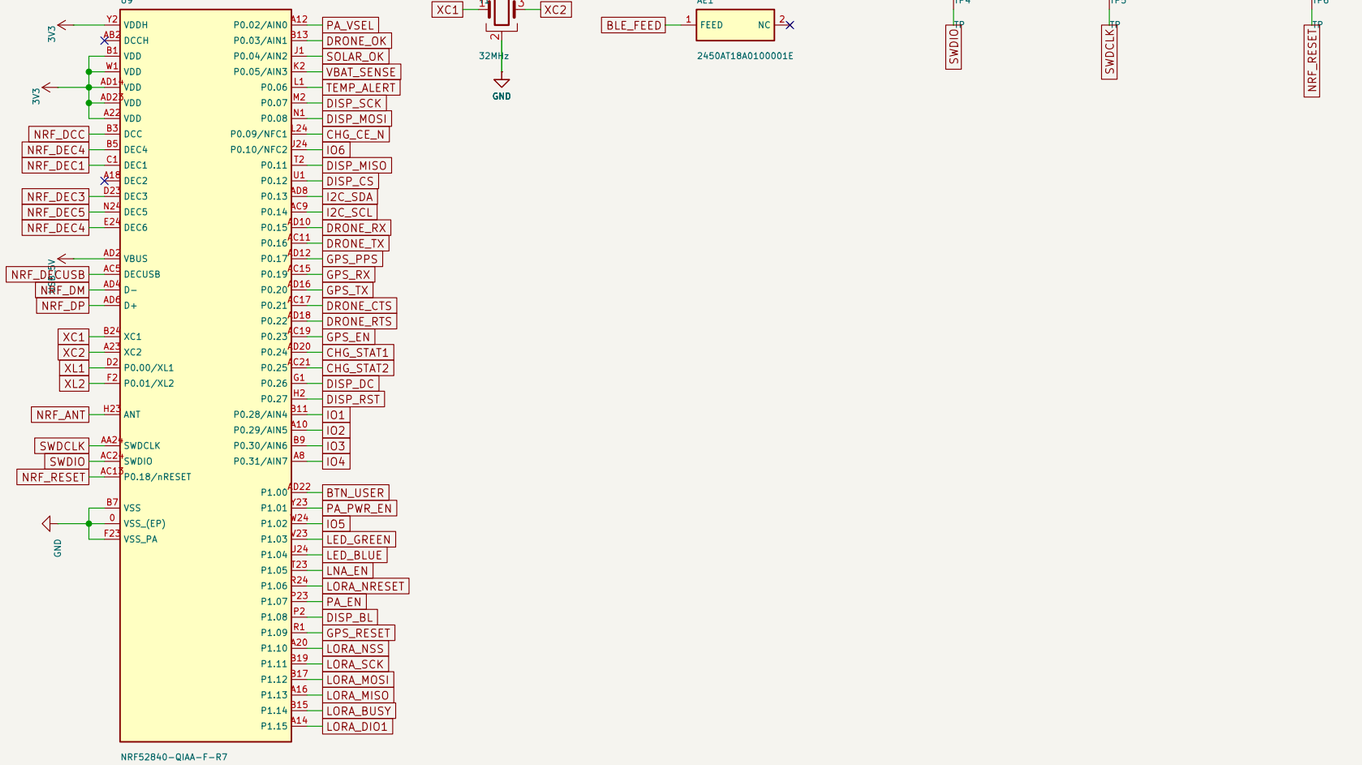
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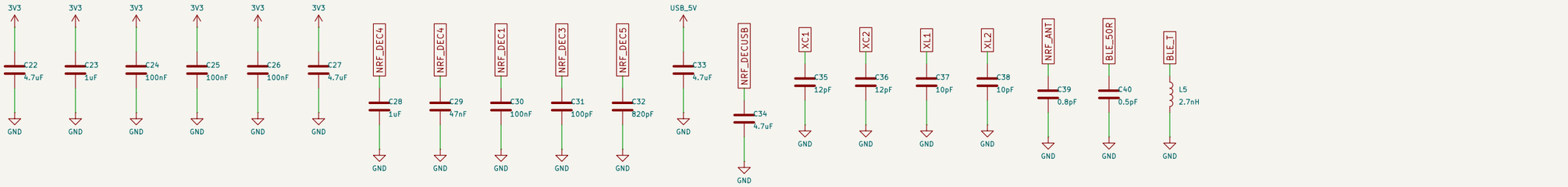
Title: System power: on/off switch, 3.3 V rail

Size: A3	Date:	**Rev: v1**
KiCad E.D.A. 10.0.6		Id: 4/9

Processor and Bluetooth: nRF52840

nRF52840 in normal-voltage mode: VDD = VDDH = the 3.3 V rail, so its pins and the SX1262 share one logic level. REG1 DC/DC on (10 uH + 15 nH). Nordic reference circuit, aQFN73.







OffGrid Devices

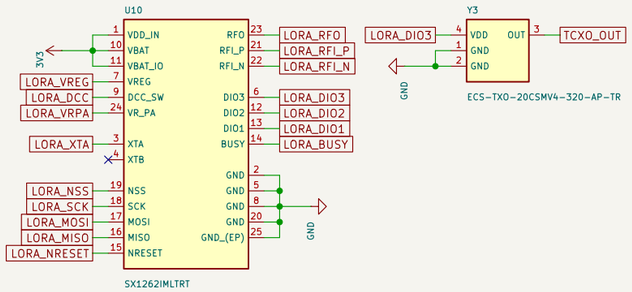
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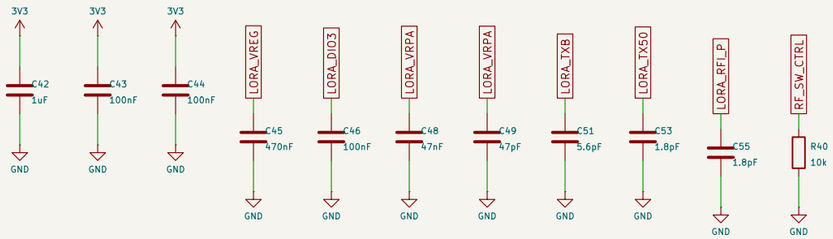
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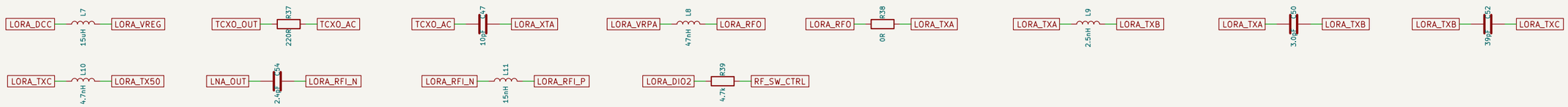
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KiCad E.D.A. 10.0.6		Id: 5/9

LoRa radio: SX1262, its clock, transmit match and receive balun

Semtech SX1262MB2CAS (915 MHz) reference values. TCXO on DIO3 (set 2.2 V in firmware), DIO2 drives the antenna switch.







OffGrid Devices

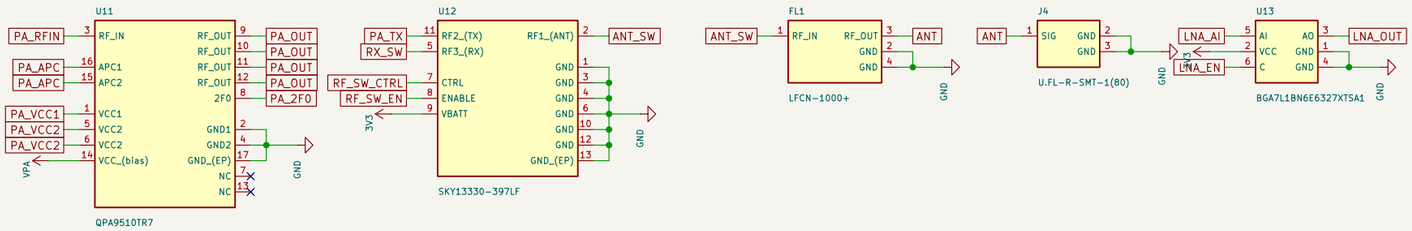
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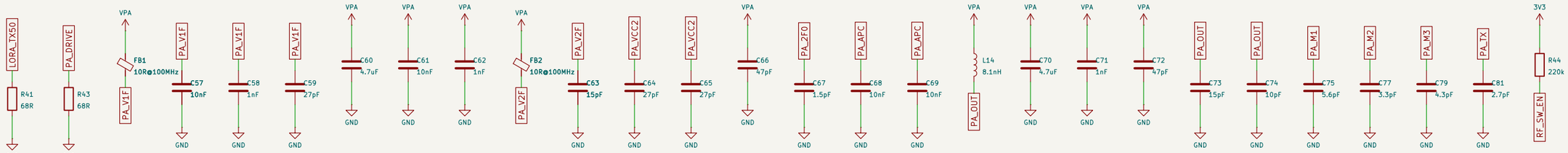
Title: LoRa radio: SX1262, its clock, transmit match and receive balun

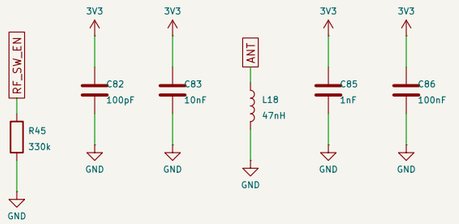
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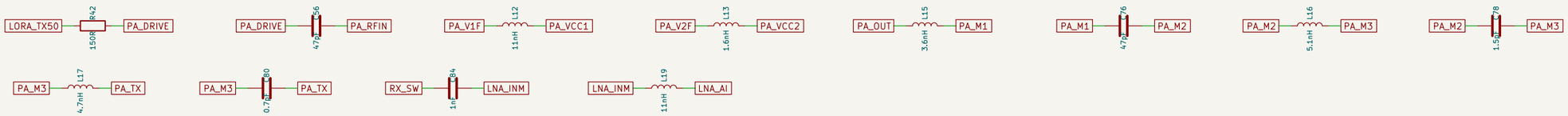
RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA

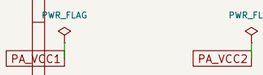
Drive: SX1262 +11 dBm -> 30 dBm (1 W), +17 dBm -> ~35 dBm (3 W). The 15.9 dB pad keeps the amplifier input below its +13 dBm limit even at the SX1262 maximum (+22 dBm -> +6 dBm).











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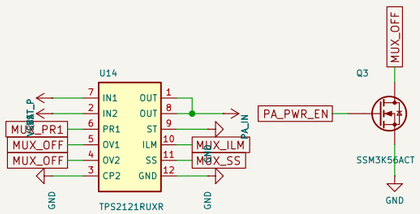
Sheet: /RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA/
File: frontend.kicad_sch

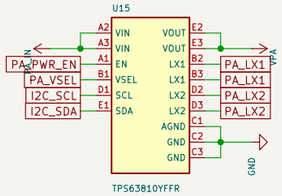
Title: RF front end: 15 dB pad, 3 W amplifier, filters, antenna switch, LNA

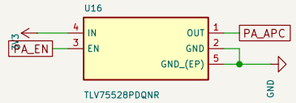
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KiCad E.D.A. 10.0.6		Id: 7/9

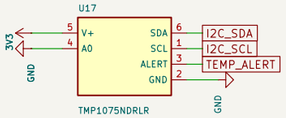
Amplifier power: battery–first supply, buck–boost, control voltage, temperature

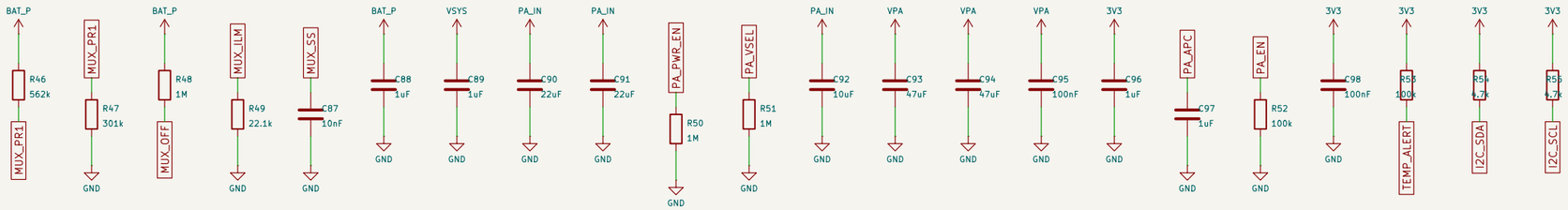
Amplifier runs from the battery through a power mux (charger SYS only when no battery), then a TPS63810 buck–boost set over I2C (0x75): ~2.7 V for 1 W, up to 4.0 V for 3 W.

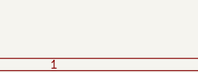












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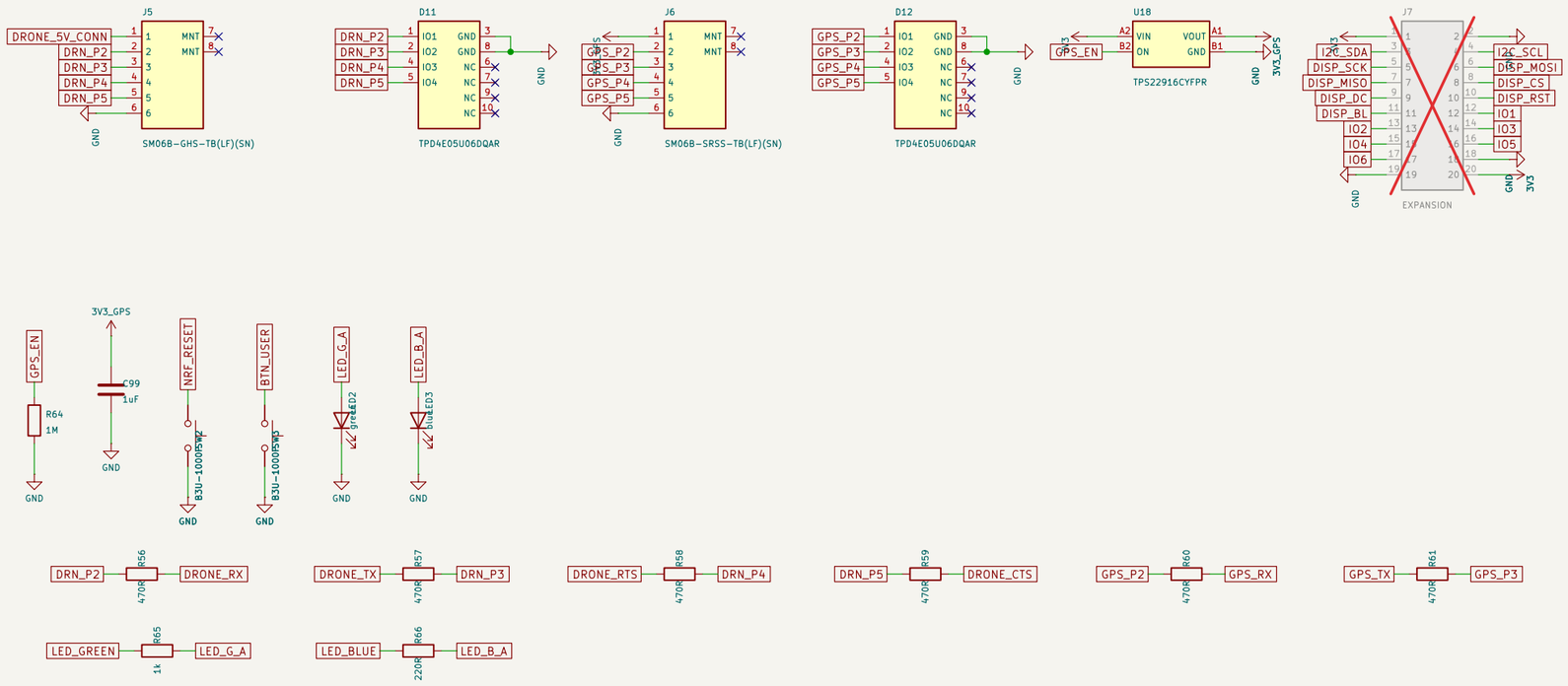
Sheet: /Amplifier power: battery–first supply, buck–boost, control voltage, temperature/
File: pa_power.kicad_sch

Title: Amplifier power: battery–first supply, buck–boost, control voltage, tem

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Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, buttons, LEDs

Drone port wired as Pixhawk TELEM (pins from the autopilot side): 1 5V, 2 TX, 3 RX, 4 CTS, 5 RTS,
6 GND. Series resistors and ESD on every external signal.



OffGrid Devices

Sheet: /Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, buttons, LEDs/
File: ports.kicad_sch

Title: Ports: drone (JST-GH), GPS add-on (JST-SH), expansion header, butto

Size: A3	Date:	**Rev: v1**
KiCad E.D.A. 10.0.6		Id: 9/9